```
import os
import h5py
import time
import matplotlib
import numpy as np
import pandas as pd
import seaborn as sns
from pandas import DataFrame
import matplotlib.pyplot as plt
from matplotlib import gridspec
import os
import h5py
import time
import matplotlib
import matplotlib.ticker as ticker
import numpy as np
import pandas as pd
import seaborn as sns
from pandas import DataFrame
import matplotlib.pyplot as plt
from matplotlib import gridspec
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.metrics import classification_report
from sklearn.pipeline import Pipeline
%matplotlib inline
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.metrics import classification_report
from sklearn.pipeline import Pipeline
import keras
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import Dropout
from deap import algorithms, base, creator, tools
import tensorflow as tf
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, Masking,
TimeDistributed
from keras.models import Sequential
from keras.layers import Dense, Conv1D
```

```
from keras.layers import Dropout, Input
from keras import initializers
from keras.layers import MaxPooling1D
from keras.models import Model
from keras.layers import TimeDistributed, Flatten
from keras.layers import Dense, Activation, Dropout
from keras.callbacks import EarlyStopping
from keras.callbacks import ModelCheckpoint
%matplotlib inline

filename = 'N-CMAPSS_DS05.h5'

# Time tracking, Operation time (min):  0.003
t = time.process_time()

# Load data
with h5py.File(filename, 'r') as hdf:
        # Development set
        W_dev = np.array(hdf.get('W_dev'))             # W
        X_s_dev = np.array(hdf.get('X_s_dev'))         # X_s
        X_v_dev = np.array(hdf.get('X_v_dev'))         # X_v
        T_dev = np.array(hdf.get('T_dev'))             # T
        Y_dev = np.array(hdf.get('Y_dev'))             # RUL
        A_dev = np.array(hdf.get('A_dev'))             # Auxiliary

        # Test set
        W_test = np.array(hdf.get('W_test'))           # W
        X_s_test = np.array(hdf.get('X_s_test'))       # X_s
        X_v_test = np.array(hdf.get('X_v_test'))       # X_v
        T_test = np.array(hdf.get('T_test'))           # T
        Y_test = np.array(hdf.get('Y_test'))           # RUL
        A_test = np.array(hdf.get('A_test'))           # Auxiliary

        # Varnams
        W_var = np.array(hdf.get('W_var'))
        X_s_var = np.array(hdf.get('X_s_var'))
        X_v_var = np.array(hdf.get('X_v_var'))
        T_var = np.array(hdf.get('T_var'))
        A_var = np.array(hdf.get('A_var'))

        # from np.array to list dtype U4/U5
        W_var = list(np.array(W_var, dtype='U20'))
        X_s_var = list(np.array(X_s_var, dtype='U20'))
        X_v_var = list(np.array(X_v_var, dtype='U20'))
        T_var = list(np.array(T_var, dtype='U20'))
        A_var = list(np.array(A_var, dtype='U20'))

W = np.concatenate((W_dev, W_test), axis=0)
X_s = np.concatenate((X_s_dev, X_s_test), axis=0)
X_v = np.concatenate((X_v_dev, X_v_test), axis=0)
```

```
T = np.concatenate((T_dev, T_test), axis=0)
Y = np.concatenate((Y_dev, Y_test), axis=0)
A = np.concatenate((A_dev, A_test), axis=0)

print('')
print("Operation time (min): " , (time.process_time()-t)/60)
print('')
print ("W shape: " + str(W.shape))
print ("X_s shape: " + str(X_s.shape))
print ("X_v shape: " + str(X_v.shape))
print ("Y shape: " + str(Y.shape))
print ("T shape: " + str(T.shape))
print ("A shape: " + str(A.shape))
```

```
Operation time (min):  0.044270833333333336

W shape: (6912652, 4)
X_s shape: (6912652, 14)
X_v shape: (6912652, 14)
Y shape: (6912652, 1)
T shape: (6912652, 10)
A shape: (6912652, 4)
```

```
W = np.concatenate((W_dev, W_test), axis=0)
X_s = np.concatenate((X_s_dev, X_s_test), axis=0)
Y = np.concatenate((Y_dev, Y_test), axis=0)
A = np.concatenate((A_dev, A_test), axis=0)

X = np.concatenate((W, X_s, A, Y), axis=1)
Y=  np.concatenate((A, Y), axis=1)
Y = DataFrame(data=Y, columns=A_var+['Y'])
X = DataFrame(data=X, columns=W_var+X_s_var+A_var+['Y'])

Y.describe()
```

```
                unit         cycle            Fc            hs
Y
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   5.239374e+00  4.002810e+01  2.226115e+00  2.958030e-01
3.900481e+01
std    2.904136e+00  2.368351e+01  8.078304e-01  4.564029e-01
2.376696e+01
min    1.000000e+00  1.000000e+00  1.000000e+00  0.000000e+00
0.000000e+00
25%    2.000000e+00  2.000000e+01  2.000000e+00  0.000000e+00
1.900000e+01
50%    6.000000e+00  3.900000e+01  2.000000e+00  0.000000e+00
3.800000e+01
```

```
75%    8.000000e+00  5.900000e+01  3.000000e+00  1.000000e+00
5.800000e+01
max    1.000000e+01  1.000000e+02  3.000000e+00  1.000000e+00
9.900000e+01
```

```
Y.drop("cycle",axis=1,inplace=True)
Y.drop("Fc",axis=1,inplace=True)
Y.drop("hs",axis=1,inplace=True)
Y.describe()
```

```
                unit             Y
count  6.912652e+06  6.912652e+06
mean   5.239374e+00  3.900481e+01
std    2.904136e+00  2.376696e+01
min    1.000000e+00  0.000000e+00
25%    2.000000e+00  1.900000e+01
50%    6.000000e+00  3.800000e+01
75%    8.000000e+00  5.800000e+01
max    1.000000e+01  9.900000e+01
```

```
X.drop("cycle",axis=1,inplace=True)
X.drop("Fc",axis=1,inplace=True)
X.drop("hs",axis=1,inplace=True)
X.drop("alt",axis=1,inplace=True)
X.describe()
```

```
               Mach           TRA            T2           T24
T30  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   5.319760e-01  5.996256e+01  4.909356e+02  5.699286e+02
1.334474e+03
std    1.208898e-01  1.834395e+01  1.985279e+01  2.112542e+01
6.869315e+01
min    3.150000e-04  2.355452e+01  4.369255e+02  4.998511e+02
1.091806e+03
25%    4.393620e-01  4.631803e+01  4.741667e+02  5.546204e+02
1.287931e+03
50%    5.371380e-01  6.257768e+01  4.960515e+02  5.678197e+02
1.329839e+03
75%    6.330240e-01  7.664008e+01  5.072438e+02  5.838195e+02
1.373982e+03
max    7.492590e-01  8.876890e+01  5.343834e+02  6.341967e+02
1.540888e+03
```

```
                T48           T50           P15            P2
P21  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   1.638378e+03  1.129818e+03  1.304474e+01  1.019378e+01
```

```
1.324339e+01
std    1.239105e+02  6.251462e+01  2.883393e+00  2.421477e+00
2.927303e+00
min    1.095354e+03  7.923581e+02  6.700688e+00  5.285918e+00
6.802729e+00
25%    1.552724e+03  1.085702e+03  1.049740e+01  7.972020e+00
1.065726e+01
50%    1.647608e+03  1.120127e+03  1.338949e+01  1.060965e+01
1.359339e+01
75%    1.711727e+03  1.165307e+03  1.528072e+01  1.209400e+01
1.551342e+01
max    1.993206e+03  1.356282e+03  2.044899e+01  1.568410e+01
2.076040e+01

                P24           Ps30           P40           P50
Nf  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   1.606218e+01  2.379398e+02  2.422144e+02  1.019697e+01
1.953364e+03
std    3.450926e+00  5.919650e+01  6.003612e+01  2.758136e+00
1.873247e+02
min    8.186232e+00  9.248497e+01  9.466065e+01  4.635623e+00
1.471000e+03
25%    1.319101e+01  1.929589e+02  1.965787e+02  7.665725e+00
1.835894e+03
50%    1.617596e+01  2.256779e+02  2.301263e+02  1.051082e+01
1.997333e+03
75%    1.847904e+01  2.724938e+02  2.773639e+02  1.232305e+01
2.112084e+03
max    2.643985e+01  4.485121e+02  4.553945e+02  1.672457e+01
2.285550e+03

                Nc            Wf          unit             Y
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
mean   8.248141e+03  2.548930e+00  5.239374e+00  3.900481e+01
std    2.277543e+02  7.877672e-01  2.904136e+00  2.376696e+01
min    7.429764e+03  5.963687e-01  1.000000e+00  0.000000e+00
25%    8.094517e+03  1.980253e+00  2.000000e+00  1.900000e+01
50%    8.237001e+03  2.340848e+00  6.000000e+00  3.800000e+01
75%    8.383625e+03  2.948869e+00  8.000000e+00  5.800000e+01
max    8.921408e+03  5.607345e+00  1.000000e+01  9.900000e+01

df_A = DataFrame(data=A, columns=A_var)
df_A.describe()

               unit         cycle            Fc            hs
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
mean   5.239374e+00  4.002810e+01  2.226115e+00  2.958030e-01
std    2.904136e+00  2.368351e+01  8.078304e-01  4.564029e-01
```

```
min    1.000000e+00  1.000000e+00  1.000000e+00  0.000000e+00
25%    2.000000e+00  2.000000e+01  2.000000e+00  0.000000e+00
50%    6.000000e+00  3.900000e+01  2.000000e+00  0.000000e+00
75%    8.000000e+00  5.900000e+01  3.000000e+00  1.000000e+00
max    1.000000e+01  1.000000e+02  3.000000e+00  1.000000e+00
```

```
print('Engine units in df: ', np.unique(X['unit']))
print('Engine units in df: ', np.unique(Y['unit']))
```

```
Engine units in df:  [ 1.  2.  3.  4.  5.  6.  7.  8.  9. 10.]
Engine units in df:  [ 1.  2.  3.  4.  5.  6.  7.  8.  9. 10.]
```

```
X_train= X.loc[(X.unit == 1) | (X.unit == 3) | (X.unit == 5) | (X.unit
== 6) | (X.unit == 7) | (X.unit == 9) | (X.unit == 10) ]
Y_train=Y.loc[(Y.unit == 1) | (Y.unit == 3) | (Y.unit == 5) |  (Y.unit
== 6) | (Y.unit == 7) | (Y.unit == 9) | (Y.unit == 10) ]
X_test=X.loc[(X.unit == 2) | (X.unit == 4) | (X.unit == 8) ]
Y_test=Y.loc[(Y.unit == 2) | (Y.unit == 4) | (Y.unit == 8) ]
```

```
print('Engine units in df: ', np.unique(X_train['unit']))
print('Engine units in df: ', np.unique(Y_train['unit']))
print('Engine units in df: ', np.unique(X_test['unit']))
print('Engine units in df: ', np.unique(Y_test['unit']))
print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)
```

```
Engine units in df:  [ 1.  3.  5.  6.  7.  9. 10.]
Engine units in df:  [ 1.  3.  5.  6.  7.  9. 10.]
Engine units in df:  [2. 4. 8.]
Engine units in df:  [2. 4. 8.]
X_train size is  (4900033, 19)
X_test size is  (2012619, 19)
y_train size is  (4900033, 2)
y_test size is  (2012619, 2)
```

```
X_train.drop("unit",axis=1,inplace=True)
X_test.drop("unit",axis=1,inplace=True)
X_train.drop("Y",axis=1,inplace=True)
X_test.drop("Y",axis=1,inplace=True)
Y_train.drop("unit",axis=1,inplace=True)
Y_test.drop("unit",axis=1,inplace=True)
print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)
```

```
X_train size is  (4900033, 17)
X_test size is  (2012619, 17)
```

```
y_train size is  (4900033, 1)
y_test size is  (2012619, 1)

C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_train.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_test.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:3:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_train.drop("Y",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:4:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_test.drop("Y",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:5:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  Y_train.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_33320\816249269.py:6:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
```

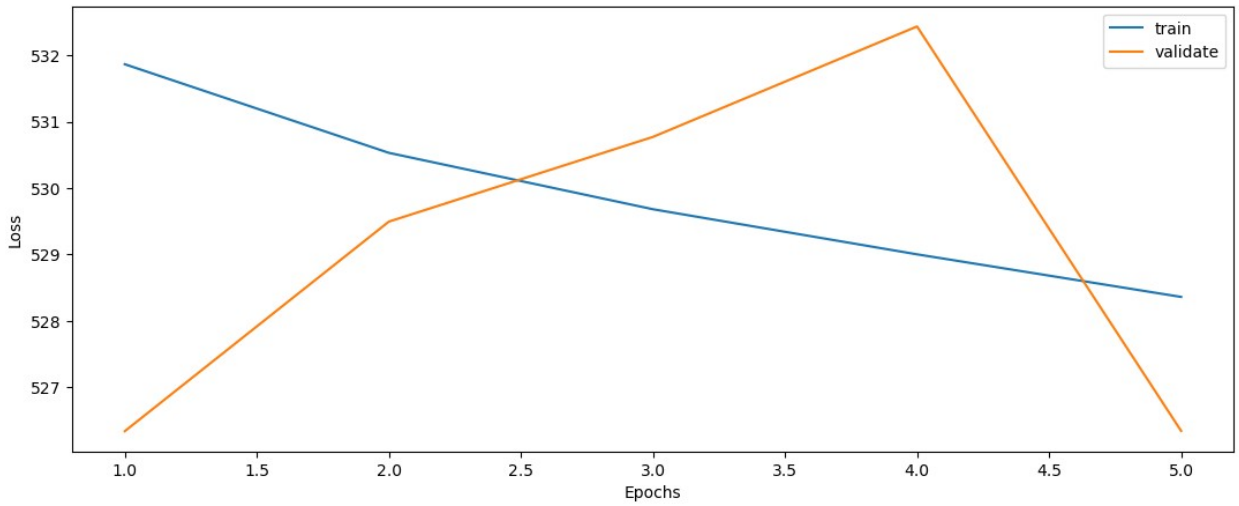
```
returning-a-view-versus-a-copy
  Y_test.drop("unit",axis=1,inplace=True)

def reshape_time_series(data, sequence_length):
    """
    Reshapes time series data for CNN input.

    Parameters:
    data (np.array): The time series dataset, expected to be a 2D
array.
    sequence_length (int): The length of the input sequences for the
CNN.

    Returns:
    np.array: Reshaped dataset suitable for CNN input.
    """
    X = []
    for i in range(len(data) - sequence_length):
        X.append(data[i:i + sequence_length])

    return np.array(X)

X_train = reshape_time_series(X_train,20)
X_test = reshape_time_series(X_test,20)
Y_train = reshape_time_series(Y_train,20)
Y_test = reshape_time_series(Y_test,20)

from keras.models import Sequential
from keras.layers import Conv1D, MaxPooling1D
from keras.layers import LSTM
from keras.layers import Dense, Dropout, Flatten

print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)

X_train size is  (4900013, 20, 17)
X_test size is  (2012599, 20, 17)
y_train size is  (4900013, 20, 1)
y_test size is  (2012599, 20, 1)

from sklearn.model_selection import train_test_split

def split_dataset(X, y, train_size=0.8):
    """
    Splits the dataset into training and validation sets.

    Parameters:
    X (np.array): Features of the dataset.
    y (np.array): Labels or targets of the dataset.
```

```
    train_size (float): Proportion of the dataset to include in the
training set.

    Returns:
    X_train, X_val, y_train, y_val: Split dataset into training and
validation sets.
    """
    X_train, X_val, y_train, y_val = train_test_split(X, y,
train_size=train_size, random_state=42)
    return X_train, X_val, y_train, y_val

X_train, X_val, Y_train, Y_val = split_dataset(X_train, Y_train,
train_size=0.8)

print("X_train shape:", X_train.shape)
print("y_train shape:", Y_train.shape)
print("X_val shape:", X_val.shape)
print("y_val shape:", Y_val.shape)
```

```
X_train shape: (3920010, 20, 17)
y_train shape: (3920010, 20, 1)
X_val shape: (980003, 20, 17)
y_val shape: (980003, 20, 1)
```

```
def reshape_labels_for_regression(y_data):
    """
    Reshapes the labels for a regression task.

    Parameters:
    y_data (np.array): Original labels with shape (number_of_samples,
20, 10).

    Returns:
    np.array: Reshaped labels with shape (number_of_samples, 1).
    """
    # Assuming we want to predict the first value at the last time
step of each sequence
    # Adjust the index '0' if a different value is desired
    y_reshaped = y_data[:, -1, 0]  # Select the last time step, first
value
    return y_reshaped.reshape(-1, 1)

model = Sequential()
model.add(Masking(mask_value=-99., input_shape=(20, 17)))
model.add(Conv1D(128, kernel_size=3, padding = "same",
activation="relu",input_shape=(20, 17)))
model.add(Dropout(0.12))
model.add(MaxPooling1D(pool_size=2, padding='same'))
model.add(Conv1D(128,kernel_size=3, padding = "same",
activation="relu"))
model.add(Dropout(0.12))
```

```
model.add(MaxPooling1D(pool_size=2, padding='same'))
model.add(TimeDistributed(Flatten()))
model.add(LSTM(units = 128, return_sequences=True))
model.add(LSTM(units = 128, return_sequences=False,
activation='tanh'))
model.add(Dense(200, activation='relu'))
model.add(Dropout(0.2))
model.add(Dense(1, activation = 'linear'))
model.compile(loss='mean_squared_error', optimizer='adam')
model.save_weights('simple_lstm_weights.h5')
es = EarlyStopping(monitor='val_loss', mode='min', verbose=1,
patience=5)
history = model.fit(X_train, Y_train,validation_data=(X_val, Y_val),
                    epochs=5,
                   batch_size=32)

Epoch 1/5
122501/122501 [==============================] - 1473s 12ms/step -
loss: 531.8674 - val_loss: 526.3387
Epoch 2/5
122501/122501 [==============================] - 1544s 13ms/step -
loss: 530.5324 - val_loss: 529.4962
Epoch 3/5
122501/122501 [==============================] - 1544s 13ms/step -
loss: 529.6827 - val_loss: 530.7727
Epoch 4/5
122501/122501 [==============================] - 1344s 11ms/step -
loss: 529.0031 - val_loss: 532.4366
Epoch 5/5
122501/122501 [==============================] - 1382s 11ms/step -
loss: 528.3615 - val_loss: 526.3427

def plot_loss(fit_history):
    plt.figure(figsize=(13,5))
    plt.plot(range(1, len(fit_history.history['loss'])+1),
fit_history.history['loss'], label='train')
    plt.plot(range(1, len(fit_history.history['val_loss'])+1),
fit_history.history['val_loss'], label='validate')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.show()

plot_loss(history)
```




```
def evaluate(y_true, y_hat, label='test'):
    mse = mean_squared_error(y_true, y_hat)
    rmse = np.sqrt(mse)
    variance = r2_score(y_true, y_hat)
    print('{} set RMSE:{}, R2:{}'.format(label, rmse, variance))


y_hat_test = model.predict(X_test)
print (y_hat_test)
```

```
62894/62894 [==============================] - 287s 5ms/step
[[36.85976]
 [36.85976]
 [36.85976]
 ...
 [36.85976]
 [36.85976]
 [36.85976]]
```

```
---------------------------------------------------------------------------
-----
ValueError                                Traceback (most recent call
last)
c:\Users\Kamal\Documents\AUC\thesiswork\N-Cmapps\Trial5.ipynb Cell 24
line 3
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=0'>1</a> y_hat_test =
model.predict(X_test)
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=1'>2</a> print
(y_hat_test)
----> <a
```

```
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=2'>3</a> evaluate(Y_test,
y_hat_test)

c:\Users\Kamal\Documents\AUC\thesiswork\N-Cmapps\Trial5.ipynb Cell 24
line 2
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=0'>1</a> def
evaluate(y_true, y_hat, label='test'):
----> <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=1'>2</a>     mse =
mean_squared_error(y_true, y_hat)
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=2'>3</a>     rmse =
np.sqrt(mse)
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=3'>4</a>     variance =
r2_score(y_true, y_hat)

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
metrics\_regression.py:442, in mean_squared_error(y_true, y_pred,
sample_weight, multioutput, squared)
    382 def mean_squared_error(
    383     y_true, y_pred, *, sample_weight=None,
multioutput="uniform_average", squared=True
    384 ):
    385     """Mean squared error regression loss.
    386
    387     Read more in the :ref:`User Guide <mean_squared_error>`.
   (...)
    440     0.825...
    441     """
--> 442     y_type, y_true, y_pred, multioutput = _check_reg_targets(
    443         y_true, y_pred, multioutput
    444     )
    445     check_consistent_length(y_true, y_pred, sample_weight)
    446     output_errors = np.average((y_true - y_pred) ** 2, axis=0,
weights=sample_weight)

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
metrics\_regression.py:101, in _check_reg_targets(y_true, y_pred,
multioutput, dtype)
     67 """Check that y_true and y_pred belong to the same regression
task.
     68
     69 Parameters
```

```
   (...)
     98     correct keyword.
     99 """
    100 check_consistent_length(y_true, y_pred)
--> 101 y_true = check_array(y_true, ensure_2d=False, dtype=dtype)
    102 y_pred = check_array(y_pred, ensure_2d=False, dtype=dtype)
    104 if y_true.ndim == 1:

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
utils\validation.py:915, in check_array(array, accept_sparse,
accept_large_sparse, dtype, order, copy, force_all_finite, ensure_2d,
allow_nd, ensure_min_samples, ensure_min_features, estimator,
input_name)
    910     raise ValueError(
    911         "dtype='numeric' is not compatible with arrays of
bytes/strings."
    912         "Convert your data to numeric values explicitly
instead."
    913     )
    914 if not allow_nd and array.ndim >= 3:
--> 915     raise ValueError(
    916         "Found array with dim %d. %s expected <= 2."
    917         % (array.ndim, estimator_name)
    918     )
    920 if force_all_finite:
    921     _assert_all_finite(
    922         array,
    923         input_name=input_name,
    924         estimator_name=estimator_name,
    925         allow_nan=force_all_finite == "allow-nan",
    926     )

ValueError: Found array with dim 3. None expected <= 2.
```

```
print (y_hat_test.shape)
print (Y_test.shape)
reshaped_data = Y_test[:, 0, 0]
reshaped_data = reshaped_data[:, np.newaxis]
print(reshaped_data.shape)
evaluate(reshaped_data, y_hat_test)
```

```
(2012599, 1)
(2012599, 20, 1)
(2012599, 1)
test set RMSE:26.011789063409275, R2:-0.05579286571155184
```

```
df = pd.DataFrame(reshaped_data)
```

```
df.to_csv('output.csv', index=False)
```